

Lecture Notes	Chapter	Folder 01 Skripten	Folder Beispiele	02	Remarks
PART I	Course Outline				
PART II	Review of FEM				
			01 Halterung		Workbench, Apdl
1	Mathematical Fundamentals				
1.1.	Differential Formulation		02 Stab		
1.2.	Variational Formulation				
1.3.	Galerkin Formulation				
1.4.	Piecewise Linear Finite Element Space				
2	Comparison of Approximation Techniques				
2.1.	Galerkin Method		03 Vergleich Galerkin, FEM, FDM		Matlab, Apdl
2.2.	FEM using Apdl				
2.3.	Finite Difference Method	Overview of Mathematical Formulations			
3	FEM in Structural Mechanics				
3.1.	Definitions				
3.2.	Tensor Algebra				
3.3.	Weak Form of Elastostatic Problems				
3.4.	Strong Form of Elastostatic Problems				
3.5.	Displacement based FEM	BAR2	04 Isoparametrisches Stabelement		Matlab, Apdl, Gauss quadrature
		Equilibrium FEM	05 Kragplatte		Workbench, Apdl
		PLANE4	06 Isoparametrisches 2D-Element		Matlab
4	FEM in Practical Use				
4.1.	Quality Control in FEA				
4.2.	Assessment of System Response				
4.3.	Guidelines for Reporting FEA				
4.4.	Commercial FE Packages				

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PART III	Modeling Aspects				
	5 FE for Structural Analysis				
	5.1. Beam and Link Elements		07 I-Träger		Apdl
	5.2. Continuum Stress Elements		08 ESZ EFZ		Apdl
	5.3. Shell Elements				
	5.4. Miscellaneous Elements				
	5.5. FE for Structural Analysis in ANSYS				
	6 Modeling Tools				
	6.1. Coordinate Systems in FEA		09 Koordinatensysteme		Apdl
	6.2. Symmetry and Antisymmetry Boundary Conditions				

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PART I	Course Outline				
PART II	Review of FEM				
			<i>00 Einführungsbeispiel</i>		Workbench, Apdl
3.5.	Displacement based FEM	BEAM2	<i>01 BEAM2</i>		Matlab, Apdl
PART IV	Dynamic Analysis				
7	Fundamentals on Dynamic Analysis		<i>02 SDOF GUI</i>		Matlab
7.1.	Free Vibration				
7.2.	Forced Vibrations				
7.3.	Numerical Examples		<i>03 SDOF Matlab</i>		Matlab
			<i>04 SDOF Workbench</i>		Workbench
			<i>05 SDOF APDL</i>		Apdl
8	Mode Superposition Method		<i>06 FEM Matlab 07 FEM Apdl 08 FEM Workbench</i>		Matlab Apdl Workbench
8.1.	Modal Transformation				
8.2.	Eigenvalue Problems				
9	Transient Analysis		<i>06 FEM Matlab 07 FEM Apdl 08 FEM Workbench</i>		Matlab Apdl Workbench
9.1.	Newmark Method				
9.2.	Full Solution Method				
9.3.	Mode Superposition Method				
10	Harmonic Analysis		<i>06 FEM Matlab 07 FEM Apdl 08 FEM Workbench</i>		Matlab Apdl Workbench
10.1.	Full Solution Method				
10.2.	Mode Superposition Method				